

RoadAssist Bharat — Review-Based Work Plan

4 members · 3 reviews · 18 weeks — "One Platform. Every Vehicle. Every Phone. Every Road."

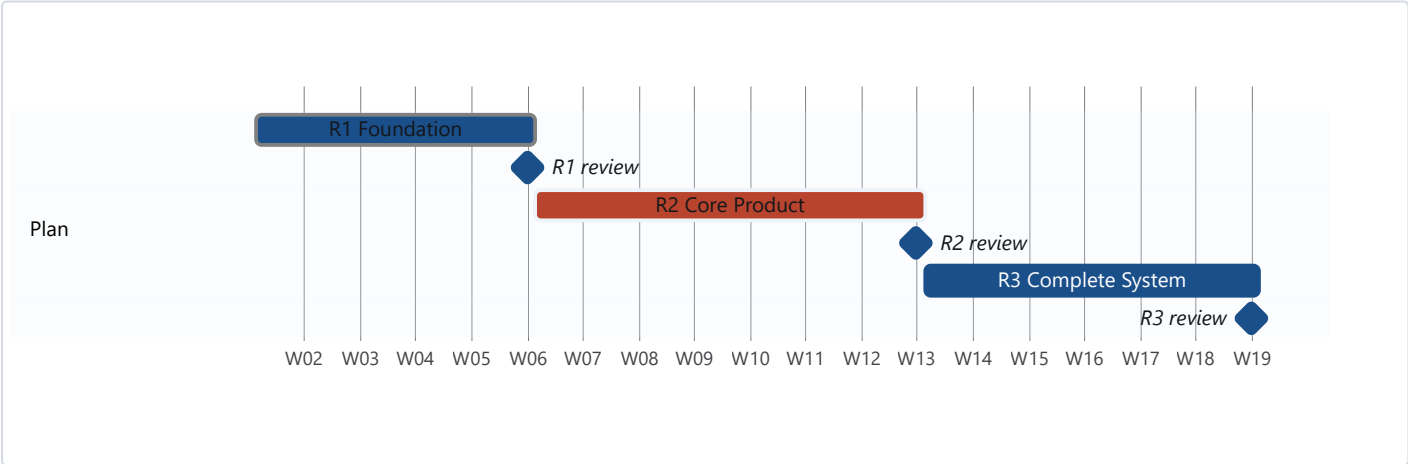
1. The Team

	Role	Owns end-to-end	Total effort
P1	Backend & Database Lead	Server services, PostgreSQL/PostGIS schema, all API contracts, authentication + RBAC, booking & dispatch engine, emergency core, payments, sync server	87 days
P2	Frontend & Mobile Lead	React Native app, web + government + admin portals, design system, offline client, maps UI, accessibility, 8 languages	86 days
P3	AI & Data Lead	9 AI subsystems, model training + serving, on-device inference, feature store, voice/NLU, analytics pipeline	70 days
P4	DevOps, QA & Security Lead	Cloud infrastructure, CI/CD, observability, all testing, security, SMS/IVR/USSD telecom gateway, deployment, compliance	76 days

Backup pairs (so no single point of failure): P1 ↔ P3 · P2 ↔ P4 Each member reviews the other's code. Nobody merges their own work.

2. Review Calendar

Review	Weeks	Duration	Theme	Completion
Review 1	1 – 5	5 weeks	Foundation — research, design, database, login working	25%
Review 2	6 – 12	7 weeks	Core Product — full booking journey, AI, offline, maps	70%
Review 3	13 – 18	6 weeks	Complete System — emergency, government, testing, deployment	100%



3. Effort Distribution

Member	Review 1 (5 wk)	Review 2 (7 wk)	Review 3 (6 wk)	Total
P1 Backend	25 days	33 days	29 days	87
P2 Frontend	17 days	36 days	33 days	86
P3 AI	13 days	33 days	24 days	70
P4 DevOps/QA	17 days	27 days	32 days	76

Member	Review 1 (5 wk)	Review 2 (7 wk)	Review 3 (6 wk)	Total
Team total	72 days	129 days	118 days	319
Per person per week	3.6 days	4.6 days	4.9 days	

Review 1 is deliberately light. Design work that is rushed costs three times as much to undo later. The load rises as the design stabilises.

Two rebalancing decisions were made so nobody is overloaded: 1. P4 takes the **Admin console and Fleet dashboard** off P2 in Review 2 (P2 would otherwise be at 6.3 days/week). 2. P1, P2 and P3 each absorb **2 days of test execution** from P4 in Review 3 (P4 would otherwise be at 6.0 days/week).

P2 remains the tightest at 5.1 days/week in Review 2. Mitigation: the **Android Auto prototype is the declared first cut** — dropping it releases 2 days without touching any graded deliverable.

REVIEW 1 — Foundation

Weeks 1–5 · 72 team-days · 25% complete

Goal: prove the problem is real, the design is sound, and the system's skeleton runs. No feature is complete yet — and that is correct at this stage.

What the panel will see

- Field evidence** — 20 vehicle owners, 10 mechanics, 2 fleet managers interviewed. Real quotes, not assumptions.
- Live database** — migrations run on an empty database, 100,000 realistic rows seeded in under 2 minutes, geospatial "nearest mechanic" query returning in under 50 ms.
- Working login** — OTP arrives on a real phone, user logs in, token refreshes automatically. Then a stolen token is replayed and the system revokes the entire session family.
- Clickable prototype** — the complete breakdown journey as a click-through, in Hindi.
- CI pipeline** — a pull request with a bug, a failing test and a hardcoded password is blocked automatically in under 8 minutes.

Work split

	P1 — Backend	P2 — Frontend	P3 — AI	P4 — DevOps/QA
Weeks 1–2	Integration feasibility study (VAHAN, ERSS 112, UPI); candidate entity list; backend backlog; monorepo scaffold	Device + connectivity study; 20 user interviews; 6 personas; 4 journey maps; wireframes for 12 screens	Data availability audit; model feasibility (which of the 9 are real in v1); labelling plan	Cloud + compliance research; start TRAI DLT registration (longest lead item); CI skeleton
Weeks 3–4	C4 architecture + 15 ADRs ; API style guide; 60-table schema , indexes, audit log, soft delete	Information architecture; design system foundations; app scaffolds; local (offline) schema	AI Gateway design; feature store schema; label capture design	STRIDE threat model; SLO definitions; database infrastructure + backups
Week 5	Authentication — OTP, JWT + rotating refresh, RBAC engine, consent ledger	Auth screens, secure token storage, silent refresh, consent centre	Fraud signal catalogue	Vault secrets, SMS gateway with 2 vendors, edge rate limiting, WAF
Days	25	17	13	17

Acceptance criteria

#	Criterion	Owner
1	Every product assumption is backed by field evidence or explicitly marked as an open bet	All
2	Migrations apply and roll back cleanly on an empty database, twice in a row	P1
3	Every table has a primary key, timestamps, soft delete, version column and an owning service	P1
4	Zero sequential scans on the top-20 queries at 100,000 rows	P1
5	OTP brute force blocked at 5 attempts / 15 minutes; refresh token reuse revokes the whole family	P1

#	Criterion	Owner
6	20 RBAC policy tests pass, including 6 negative access-control cases	P1
7	Wireframes cover loading, empty, error and offline states for all 12 screens	P2
8	Design system renders correctly in all 8 languages with no text truncation	P2
9	Each of the 9 AI systems is classified honestly: trainable now / trainable later / rules-only in v1	P3
10	Every model has a baseline recorded before training begins	P3
11	CI blocks lint errors, failing tests and secrets in under 8 minutes	P4
12	Every regulatory obligation is mapped to the phase that satisfies it	P4

REVIEW 2 — Core Product

Weeks 6–12 · 129 team-days · 70% complete

Goal: a person can use this. The complete breakdown journey works on a real phone, on a real network — and keeps working when the network disappears.

What the panel will see

- A full booking, end to end** — request help → AI diagnosis → 3 mechanics offered → one accepts → live tracking on a map → work completed → UPI payment → rating.
- The offline demo (the differentiator)** — phone in **airplane mode**: add a vehicle, run a diagnosis using the on-device model, create a help request, view an offline map. Reconnect. Everything syncs, one deliberate conflict is raised and resolved.
- The feature-phone demo** — on an actual ₹1,200 phone with no internet: send **MADAD** by SMS, receive the reply, confirm location, get the mechanic's details, cancel. Then the same in Tamil over an IVR call.
- AI with the plug pulled** — the AI service is switched off completely, mid-demo. The product keeps working on rule-based fallbacks. Nothing breaks.
- A real diagnosis** — photograph an actual damaged part, get the fault, severity, parts list and a "do not drive this" verdict in Hindi.

Work split

	P1 — Backend	P2 — Frontend	P3 — AI	P4 — DevOps/QA
Weeks 6–8	140 API endpoints ; booking state machine; dispatch engine; pricing; UPI payments; webhooks	Design system + 48 screens ; onboarding, garage, request-help flow, live tracking, payment	AI Gateway live with rule-based logic for all 8 capabilities (so nothing waits on models)	Observability stack (metrics, logs, traces); preview environments; mobile CI with signed builds
Weeks 9–10	AI client with circuit breakers + fallback registry ; feature pipeline; BFF composite endpoints	Mechanic app; web PWA; AI confidence UI; user feedback loop	7 models trained — diagnosis, image, voice, maintenance, matching, crash, fraud	ML infrastructure; GPU cost controls; Admin console + Fleet dashboard (taken off P2)
Weeks 11–12	Sync server — change feed, conflict resolution, tombstones, replay; geofencing; diagnostics service	Offline client — local DB, operation queue, sync engine, conflict UI; maps + offline packs; OBD Bluetooth	On-device model bundle ≤ 15 MB ; vehicle-class rule packs; EV diagnostics	Telecom gateway — SMS commands, IVR in 8 languages, USSD; tile server; network chaos harness
Days	33	36	33	27

Acceptance criteria

#	Criterion	Owner
1	100% of the v1 API implemented; specification generated from code, never hand-written	P1
2	Every write endpoint is idempotent — proven by a replay test	P1

#	Criterion	Owner
3	The booking state machine rejects every illegal transition (exhaustive test)	P1
4	Complete booking journey works on Android 10 / 2 GB RAM / throttled 3G	P2
5	Every screen has loading, empty, error and offline states built — verified in review	P2
6	Airplane-mode journey completes and syncs with zero data loss	P2 + P1
7	30-scenario conflict test suite passes; no scenario produces a corrupt state	P1 + P2
8	Every model beats its recorded baseline on a held-out test set	P3
9	On-device inference under 120 ms on the low-end reference phone	P3
10	Diagnosis never wrongly says "safe to drive" — measured false-safe rate is zero	P3
11	Kill the AI service entirely: the full booking still completes on fallbacks	P1 + P3
12	A feature-phone user can request help, track and cancel entirely over SMS	P4
13	Navigation and search work with the device fully offline	P2 + P4
14	Sync of 100 queued operations completes in under 8 seconds on 3G	P1 + P2

REVIEW 3 — Complete System

Weeks 13–18 · 118 team-days · 100% complete

Goal: the life-safety features work, the system is tested under failure, and it is deployed and running in production.

What the panel will see

- Crash to responder** — a simulated crash on a real phone. A 30-second countdown takes over the locked screen with alarm and vibration. It completes. Emergency contacts receive an SMS with a live location link, the nearest responder is notified, the incident room opens. **Under 10 seconds.**
- The proof that it is engineered, not assembled** — the entire platform is scaled to **zero replicas**, live, in front of the panel. An SOS is triggered. It still works, over the isolated degraded path.
- False-positive proof** — driving over a real speed breaker on camera. Nothing happens. Measured over 200 hours of real driving.
- Government dashboard** — log in as a district officer, see live incidents in that jurisdiction only, identify an accident blackspot, generate the statutory monthly report as a PDF. Then **attempt to identify an individual citizen and fail** — with the audit log showing the attempt.
- Accessibility proof** — the complete booking journey driven by a screen reader **with the display switched off**.
- Deployment** — one commit reaches production through the full pipeline. Then a broken version is deployed on purpose and rolls back automatically in under 2 minutes.

Work split

	P1 — Backend	P2 — Frontend	P3 — AI	P4 — DevOps/QA
Weeks 13–14	Emergency engine — escalation ladder, ERSS 112 handoff, break-glass medical access, degraded SMS path	SOS UI + crash countdown + incident room; government portal with live map and 6 statutory reports	Crash detection tuned to production; severity model; blackspot detection; incident forecasting	Isolated emergency infrastructure; chaos test proving SOS survives total platform failure ; mTLS for government
Weeks 15–16	Anonymization with k-anonymity ≥ 10 ; CDC pipeline; property-based + mutation testing	Analytics dashboards; fleet + mechanic screens; Detox end-to-end tests ; device farm; accessibility + 8-language QA	ClickHouse analytics + metric dictionary ; bias audit across all models; adversarial testing	Load testing (10k requests/sec), chaos, security testing ; bug burn-down to zero critical issues
Weeks 17–18	Online migrations; health probes; query tuning, caching; scaling plan + tech debt register	App store submission; over-the-air updates; APK size, startup, jank and data-usage optimisation	Model serving + automated rollback; inference optimisation; AI roadmap and data flywheel	Terraform production, ArgoCD, disaster-recovery drill ; infrastructure right-sizing; cost model; compliance roadmap

	P1 — Backend	P2 — Frontend	P3 — AI	P4 — DevOps/QA
Days	29	33	24	32

Acceptance criteria

#	Criterion	Owner
1	Crash → contacts and responders notified in under 10 seconds (95th percentile)	P1
2	Emergency path survives a chaos test with every other service down	P4
3	The AI cannot auto-dispatch emergency services — enforced in code, not policy	P1 + P3
4	Crash false-positive rate ≤ 1 per 1,000 driving hours, measured over real driving	P3
5	Break-glass medical access logs the actor and reason, and notifies the user afterwards	P1
6	No citizen is identifiable in any government view or export — proven by a re-identification test	P1 + P2
7	Test coverage ≥ 80% on new code; zero open critical or high-priority bugs	P4
8	Load test: 95th percentile under 300 ms at 10,000 requests/second	P4
9	Zero high or critical security findings survive	P4
10	Full journey passes a screen-reader walkthrough (WCAG 2.1 AA)	P2
11	Disaster-recovery drill: restore into a clean region, recovery time under 4 hours	P4
12	Rollback of any service completes in under 5 minutes — proven by drill	P4
13	Every metric has exactly one definition; no two dashboards disagree	P3
14	Every alert has a runbook a new person can follow unaided	P4

4. Weekly Working Rules

When	What	Duration
Daily 9:45	Standup — <i>yesterday / today / blocked</i> . Hard stop.	12 min
Daily 16:30	Pull request sweep — every open PR reviewed or explicitly deferred	30 min
Daily 18:00	Build must be green. Whoever broke it fixes or reverts it.	—
Monday	Sprint planning (2-week sprints)	60 min
Thursday	Risk + security review	30 min
Friday	Demo of working software (recorded) + retrospective	45 min

Rules that are not negotiable 1. Branch names: `feat/r2-p1-booking-state-machine` — *type / review / owner / summary*. Maximum 3 days alive. 2. Commits: `feat(booking): reject illegal state transitions` — type, scope, imperative summary. 3. Every pull request needs **1 approval; 2 approvals** for database migrations, authentication, payments and the emergency service. 4. Nobody merges their own code. 5. Blocked for more than 2 hours? Say so publicly. Sitting silently on a blocker is the only genuinely unacceptable behaviour on this team.

Definition of Done — code merged · tests written (≥ 80%) · works on the low-end reference phone · has loading/empty/error/offline states · documented · demoed on Friday.

5. If You Fall Behind — Cut In This Order

Decide this now, not under pressure the week before a review.

Order	Cut	Days freed	Cost
1	Android Auto prototype	2	Zero — it was always a bonus
2	USSD (keep SMS + IVR)	3	Low — SMS and IVR already cover feature phones

Order	Cut	Days freed	Cost
3	Sound-based diagnosis	2	Low — it is already flagged as a spike
4	Fleet dashboard	4	Medium — reduces the business story
5	3 of the 8 languages	3	Medium — keep Hindi, English, Tamil, Telugu, Marathi
6	Predictive maintenance model	4	Medium — rules-based version still ships

Never cut, under any circumstance: the emergency path, the offline engine, the SMS fallback, or security testing. These are the four things that make this a platform rather than an app.

6. Top Risks

Risk	Who	What we do about it
TRAI DLT approval for SMS takes weeks	P4	Started in Week 1 , before anything else. Two gateway vendors registered.
No real training data at the start	P3	Every AI feature ships rule-based in Review 2. Models are an upgrade, never a dependency.
Offline sync corrupts booking data	P1 + P2	Conflict matrix written <i>before</i> coding; 30-scenario test suite; server is always authoritative on booking state.
Crash detection falsely calls emergency services	P3 + P1	Model raises a <i>signal</i> , never an incident. 30-second cancel window. Call-back verification. Auto-dispatch blocked in code.
P2 overloaded in Review 2	P2	Admin + Fleet portals moved to P4; Android Auto is the declared first cut.
Team of 4 undersized for the scope	All	Reviews are milestone-gated, features are flag-controlled, and the cut list above is agreed in advance.